

GenePath@ System

Instruction Manual

Catalog Numbers
310-0000
through
310-0017

BIO-RAD

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Warranty

The **GenePath** power module, chamber, variable speed pump, and accessories are warranted against defects in materials and workmanship for 1 year. If any defects occur in the instruments or accessories during this warranty period, Bio-Rad Laboratories will repair or replace the defective parts at its discretion without charge. The following defects, however, are specifically excluded:

Defects caused by improper operation.

Repair or modification done by anyone other than Bio-Rad Laboratories or an authorized agent.

Damage caused by substituting an alternative chamber or pump.

Use: of fittings or spare parts supplied by anyone other than Bio-Rad Laboratories.

Damage caused by accident or misuse.

Damage caused by disaster.

Corrosion caused by improper solvent* or sample.

This warranty does not apply to parts listed below:

Fuses

Tubing

Electrodes

For any inquiry or request for repair service, contact Bio-Rad Laboratories. Inform Bio-Rad of the model and serial number of your instrument.

Important: This Bio-Rad instrument is designed and certified to meet IEC1010-1* safety standards. Certified products are safe to use when operated in accordance with the instruction manual. This instrument should not be modified or altered in any way. Alteration of this instrument will:

Void the manufacturer's warranty

Void the IEC1010-1 safety certification

Create a potential safety hazard

Bio-Rad Laboratories is not responsible for any injury or damage caused by the use of this instrument for purposes other than for which it is intended or by modifications of the instrument not performed by Bio-Rad Laboratories or an authorized agent.

*IEC1010-1 is an internationally accepted electrical safety standard for laboratory instruments.

The **GenePath** chamber is not compatible with chlorinated hydrocarbons (e.g., chloroform), aromatic hydrocarbons (e.g., toluene, benzene), or acetone. Use of organic solvents voids all warranties.

Section 1

General Information

1.1 Safety

The GenePath system uses high voltage and current and should be operated with care at all times. The safety interlocks are for your protection and should not be circumvented. To avoid shock, set up the GenePath components in a dry area. Immediately wipe up any spilled buffers or salt solutions.

When pausing or aborting a run, always check that the current display goes to zero or displays off. This can take 2-5 seconds while the power supply discharges. It is then safe to open the lid of the cell.

Warning: There are high voltages and currents within the cell, which can be harmful. Do not attempt to circumvent these safety interlocks. Always turn off the power to the cell before working within the gel box.

The Cooling Module is ground isolated. Although there is virtually no current flowing through the Tygon tubing into the cooling module, you should avoid assembling or disassembling the tubing while the GenePath system is operating.

When inserting or removing the external temperature probe, be sure that the GenePath system is turned off. Do not handle the probe with wet hands.

Definition of Symbols



Caution, risk of electric shock

A ! Caution (refer to accompanying documents)

1.2 Overview

Pulsed field electrophoresis is a technique for resolving chromosome-sized DNAs. By alternating the electric field between spatially distinct pairs of electrodes, megabase (mb) sized DNAs are able to re-orient and move at different speeds through the pores in an agarose gel. Overview articles and specific applications are listed in Section 6.

The GenePath system separates large and small DNA fragments with better resolution, speed, and accuracy, than initial pulsed field methods. DNAs from 1,000 bases to over 10 megabases (mb) may be effectively resolved. Applications of the GenePath system include nosocomial testing, epidemiological testing, and pathogen strain typing.

The GenePath system is based on two leading technologies. The first is CHEF³⁰⁴ (Clamped Homogeneous Electric Fields). Like the CHEF-DR[®] III system, the GenePath system provides highly uniform, or homogeneous, electric fields within the gel. This is accomplished using an array of 24 electrodes, which are “clamped” or held to intermediate potentials to eliminate lane distortion. Thus, lanes are straight. And, like the CHEF-DR III system, the GenePath system maintains uniform fields using patented Dynamic Regulation (US Patent 4,878,008). The electrodes sense changes in local buffer conductivity due to buffer breakdown, or change in buffer type, gel thickness, or temperature. Potentials are readjusted immediately to maintain uniform fields, thus insuring high resolution during runs and between runs.

The second technology is PACE³⁰⁵ (Programmable Autonomously Controlled Electrodes), where the voltage potential of each of the 24 electrodes is regulated independently. The GenePath system has pre-programmed run conditions for a variety of common pathogens. The operator merely selects the appropriate program and the GenePath performs the optimal pulsed field switching protocol for that organism. There is also a battery backed-up RAM and automatic restart after power interruption to insure reproducible results even when the power source is unreliable.

1.3 Specifications

GenePath System

Dimensions	43 (deep) x 48 (wide) x 17.5 (high) cm
Construction	Aluminum chassis
Weight	10 kg
Total system weight	40 kg
Power supply	350 V maximum, to allow maximum gradient of 9 V/cm, continuously adjustable: built in
Electrical	
Maximum current	0.5 amperes
Allowable voltage gradients	0.6–9 V/cm, in 0.1 V/cm increments
Battery back-up	All parameters in memory
Electrode potentials	Dynamically regulated (feedback adjustment) $\pm 0.5\%$
Data entry	Keyboard
Functional	
Switching range	0.1 sec to 65K sec
Switch angle variable	90-120 degrees (all electronic switching) in 1" increments
Maximum program blocks	3, with automatic execution
Maximum run time	999 hours per block
Input voltage range	90-132 VAC/47–63 Hz/4 amps 198-264 VAC/47–63 Hz/2 amps
Fuses	0.5 amp Fast Blow for high voltage output 3.15 amp Slow-Blow (100/120 V) or 1.60 amp Slow-Blow (220/240 V)
Environmental	
Operating	50 °F (10 °C) to 90 °F (32 °C) temperature 30–80% humidity
Storage	32 °F (0 °C) to 140 °F (60 °C) temperature 10–90% humidity

Electrophoresis Cell

Dimensions	11.4 (high) x 44.2 (wide) x 50.3 (deep) cm, horizontal format
Construction	Cover: Vacuum formed polycarbonate Base: Injection molded polycarbonate
Lid	Safety interlocked
Weight	10.2 kg
Electrodes	24 platinum (0.02 inch diameter)
Temperature monitoring	Via precision temperature probe mounted in base of cell

Accessories Included

Variable speed pump	120 V, ground isolated. Flow rate 1 liter/min, typical
Casting stand	14 cm x 13 cm
Comb	15 well comb and comb holder
Tygon tubing	365 cm
Disposable sample plug mold	50 slot
Yeast DNA Standard	S. cerevisiae YNN295, 2 plugs
Chromosomal grade agarose	5 grams
Pulsed field certified agarose	5 grams
Leveling bubble	1
Fuses	0.5 Amp Fast Blow, 2
Manual	1

CoolinaModule

Weight	14kg
Construction	Aluminum
Dimensions	42 cm long x 23 cm wide x 24 cm high
Cooling capacity	75 watts of input power at 14 °C
Operating range	5 °C-25 °C
Fuse	100 & 120 V: 6.3 amp, 250 V Slo-Blow 220/240V: 3.15 Amp, 250 V Slo-Blow

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

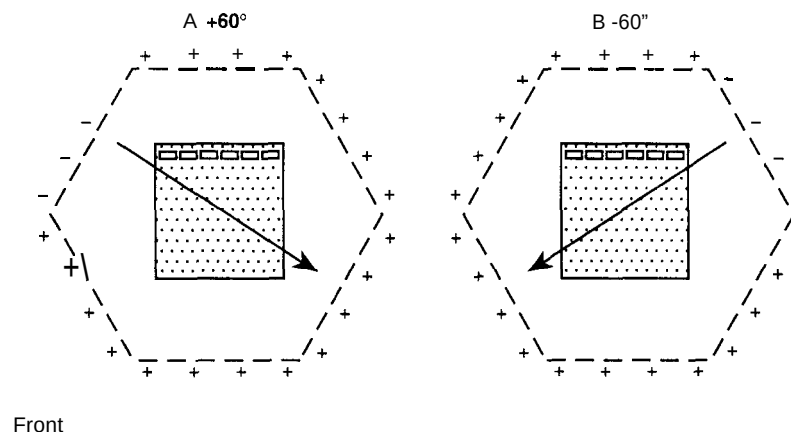
1.4 Description of Major Components

GenePathSystem

The GenePath system contains all the electronics necessary to perform pulsed field electrophoresis. It contains a 350 V power supply, the switching functions, and drivers for the 24 electrodes. The drivers provide clamped homogeneous electric fields in the electrophoresis cell, and maintain them regardless of the field angle selected. This **feature**, dynamic regulation (US Patent 4,878,008 issued to Bio-Rad Laboratories), modulates the potentials so that the proper voltages are maintained regardless of gel size, or fluctuations in buffer conductivity or temperature. The fused power supply operates with a maximum voltage gradient of 9 V/cm, or 300 V.

Figure 1.1 A shows the relative potentials of each electrode pair when the $+60^\circ$ vector (indicated by the arrow) is activated. Net field vector is from NW to SE. The highest potentials are found along the SE segment of the hexagon. The potentials gradually decline along the adjacent segments. The NW segment, directly opposite the SE, has 0 potential, represented in the diagram as negative terminals. When the -60° angle is activated, the pattern of electric charges is as shown in Figure 1.1 B. Together, the two pulses result in a 120° included field angle. Other angles will result in values for the relative electrode potentials, according to predetermined values.

Back



Front

Fig. 1.1. Voltage clamping by the GenePath system. A. Relative electrode potentials when the $+60^\circ$ field vector is activated. **B.** Relative electrode potentials when the -60° field vector is activated.

Electrophoresis Cell

The GenePath electrophoresis cell consists of a 43 x 44 cm (17" x 17.5") acrylic box with 24 horizontal electrodes arranged in a hexagon. Gels are electrophoresed horizontally, submerged under recirculated buffer. A 14 x 13 cm (5.5" x 5") gel is cast on a platform in a separate casting stand. The platform (along with the gel) is removed from the casting stand, and placed in the center of the hexagon. The platform is held in place by a frame positioned on the cell floor. A combination wide/long format is available as an accessory. DNA migration and buffer flow is in the direction of the arrow mounted on the lid.

The electrodes are individually wired to the 24 pin computer cable, which in turn connects to the GenePath system. The individual electrodes are replaceable for easy maintenance (see Section 4). Electrodes are heavy duty, 0.02" diameter platinum wire. They are each sealed with an O-ring and silicone sealant to provide double protection against leakage.

There are two small cells below the level of the main chamber floor at the front and rear of the main cell. These cells are used for buffer circulation and priming the pump. Buffer enters the main cell through six holes in the floor near the top. A flow baffle is located just in front of these holes to prevent gel movement. Buffer exits the chamber at the front through the T fitting. One arm is for draining, the other for circulation. The base of the chamber has four leveling screws for even gel submersion in buffer.

The lid contains an interlock for safety. The voltage directly passes from the GenePath system through a short-path in the lid interlock. If the lid is opened the current flow is broken and high voltage to the gel chamber is disrupted. The cell also contains an internal temperature probe, which monitors buffer temperature in the cell, and regulates cooling by the Cooling Module.

Pump and Accessories

Each system includes a variable speed pump. The pump provides a suitable flow rate of buffer through the cell. Substitution of other pumps could pose a safety hazard and cause improper flow, and therefore lower resolution. The pump's power supply is electrically isolated within the GenePath system for safety. Its voltage requirement is independent of the line voltage supplied to the power module (e.g. 120, 100,220, or 240 volts). This pump should not be plugged into any equipment other than the GenePath system.

The pump is connected to Tygon® or plastic tubing. This tubing circulates buffer in and out of the cell. The tubing may also pass through a water cooling module. In this case, the pump should be located after the cooling module, so that buffer flows through the cooling module and then to the pump. Typically, the dial is set at 70, about 1 L/min.

Cooling Module

The Cooling Module is a stand alone, portable refrigerated apparatus specifically designed for use with the GenePath system. Electrophoresis buffer is circulated by the variable speed pump directly through the unique heat exchanger, which is a tube within a tube. Buffer circulates through the inner stainless steel tube, while liquid refrigerant circulates through the outer copper tube, resulting in rapid and efficient cooling at a rate of 0.75 °C/minute (from ambient temperature to 14 °C). The cell's internal temperature probe regulates cooling by the Cooling Module, resulting in precise maintenance of buffer temperature in the electrophoresis cell.

Section 2 Operation

2.1 Instrument Setup

Place the GenePath electrophoresis cell on a level surface, with the GenePath system to the right or on a shelf above. The electrophoresis cell should be positioned with the two quick release tube connectors facing you and all the cables in the back. Place the Cooling Module to the left of the cell. Place the variable speed pump at the rear of the cell and connect the plug from the pump to the PUMP CONNECTOR port on the back of the GenePath system. Level the electrophoresis cell with the four leveling feet at each corner of the electrophoresis cell by placing the leveling bubble (provided) in the center of the base of the cell.

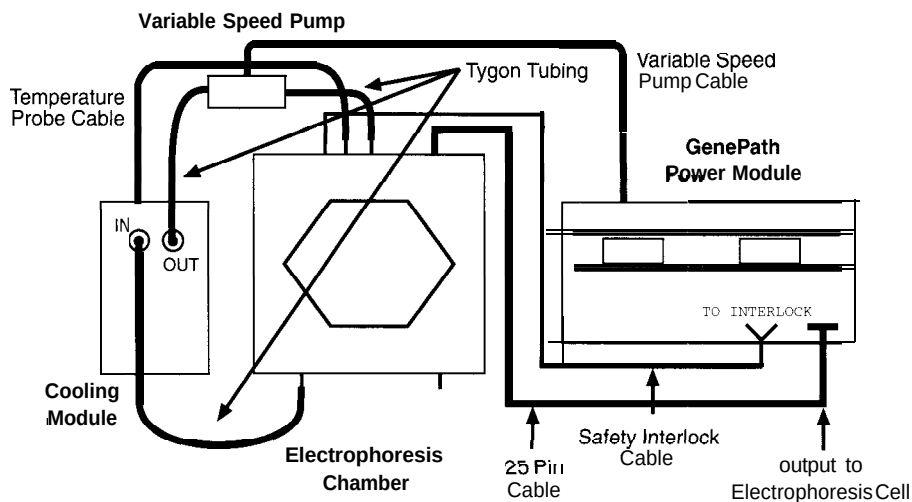


Fig. 2.1. Interconnections between components of the GenePath system.

Attach the power cords for the **GenePath** system and Cooling Module in the back of each instrument. Be sure the **GenePath** system is off. Connect the 25-pin cable from the electrophoresis cell to the OUTPUT TO ELECTROPHORESIS CELL port on the front panel of the **GenePath** system. The 25-pin cable has an safety interlock on one end, which is the end that is attached to the **GenePath** system. Connect the coiled interlock cable from the electrophoresis cell to the TO INTERLOCK jacks on the **GenePath** system.

2.2 Electrophoresis Cell Operation

To connect the Cooling Module, attach approximately 1-2 feet of 1/4 inch internal diameter Tygon tubing to both the FLOW IN and FLOW OUT ports on the Cooling Module, and secure the tubing with the plastic clamps provided. Connect 2 feet of 3/8 inch internal diameter Tygon tubing directly from the left front port of the chamber to the 1/4 inch tubing from the FLOW IN of the Cooling Module using the 3/8 to 1/4 inch reducer provided. The pump should be positioned between the outlet of the Cooling Module and the inlet of the electrophoresis cell. Connect the 1/4 inch tubing from the FLOW OUT of the Cooling Module to the inlet of the **GenePath** pump. Connect approximately 1 to 2 feet of 1/4 inch Tygon tubing from the outlet of the pump to 1 foot of 3/8 inch Tygon tubing using a 1/4 to 3/8 inch reducer. Connect the 3/8 inch tubing to the single connector at the rear of the electrophoresis cell. Finally, connect a 6 inch piece of Tygon tubing to the right front port of the chamber. This tube will be used to drain the buffer in the electrophoresis cell.

Insert the gel frame into the positioning holes in the electrophoresis cell. There are two sets of three positioning holes. Place the frame only in the holes at the bottom corners of the gel, opposite the wells, so that the frame is centered in the cell. The outermost holes are for the optional 21 x 14 cm (8.25 x 5 inch) gel format (See Casting Stand Instruction Manual for more details).

Connect the temperature probe cable from the back of the electrophoresis cell to the REMOTE SENSOR port on the back of the Cooling Module.

Establish the correct buffer flow prior to attempting any electrophoresis runs. When the desired flow rate has been achieved, subsequent electrophoresis runs should be with the same setting on the variable speed pump. Fill the cell with 2 liters of the buffer of choice. The desired flow rate of buffer through the electrophoresis cell is approximately 0.8-1 liter per minute (approximately 70 on the variable speed pump). Turn on the pump switch on the front panel of the power module.

It is also very beneficial to fine tune the Cooling Module before attempting any electrophoresis runs. Turn on the Cooling Module and pump approximately 1/2 hour before adjusting the temperature. Initially, it will be necessary to fine tune the temperature setting to achieve a consistent running temperature.

2.3 Buffer Circulation

Level the **GenePath** electrophoresis chamber by adjusting the leveling screws at the corners. Place the leveling bubble in the center of the base of the cell. Position the frame in the electrophoresis cell so that the frame is centered (center hole in each group of three holes). Pour 2 liters of buffer (appropriate concentration of TBE) into the chamber. Switch on the **GenePath** power, then switch on the variable speed pump. Circulate at -1 L/min. (a setting of -80 on the pump regulator). Maintain the flow rate at the maximum setting that does not disturb the gel. Allow the buffer to equilibrate to the desired temperature. We recommend 14 °C buffer temperature in the cell.

Before beginning the electrophoresis run, the current output displayed on the **GenePath** system should be checked to insure that the correct buffer concentration is used. The following values are for 2 liters of buffer at 14 °C circulating through the electrophoresis cell.

Organism	Current Range
<i>Staphylococcus</i>	100–135 mA
<i>Pseudomonas</i>	100–135 mA
<i>Enterococcus</i>	100–135 mA
<i>Enterobacter</i>	100–135 mA
<i>Escherichia coli</i>	100–135 mA
<i>Candida</i>	60–95 mA

If the current output is significantly different from the values listed above, the buffer should be drained, and new buffer should be added.

2.4 GenePath Operation

This section describes general operation. See manuals of the appropriate Pathogen Group Reagent Kits for sample preparation, gel casting, gel running, and staining.

Power Up

The GenePath front panel display consists of a two LED displays. The left LED displays parameters using the MENU, RUN TIME, PAUSE, START, and accompanying RAISE and LOWER keys. Select the appropriate pathogen specific program on the left LED display by pressing the MENU key and the INCREASE or DECREASE keys until the desired program is displayed. The right LED displays current in milliamperes.

Program	LED Code
<i>Staphylococcus</i>	StA
<i>Enterococcus</i>	Enc
<i>Candida</i>	CAn
<i>Pseudomonas</i>	PSU
<i>Enterobacter</i>	Enb
<i>Escherichia</i>	Eco
Screen Mode	Screen (See Section 7.1)
Download	dLd (See Section 7.2)

Run Program

When the program is selected, it may be initiated by pressing START. When the program is in progress the left LED will display the time remaining (hours) with RUN TIME lit, and the right LED will display the actual current (milliamperes) with PAUSE or START lit.

Run Time	This will display the remaining time for the program in progress.
Current	This will display the actual current for the program in progress. If the power supply is at current limit (500 mA maximum), the light will flash.
start	This will start the program and the indicator light will be lit.
Pause	Pressing this key will put the program into PAUSE and the current indicator will flash. Pressing the START key again will resume the program.

Program Termination

The program in progress may be manually terminated by holding down PAUSE for 3 to 4 seconds. When the program is terminated a double beep will sound and the right display will show **OFF**. Pressing START again will start the program from the beginning.

When the program terminates under the timer control, the START light will go off, it will sound two beeps per second for 5 seconds, and the right display will show **OFF**. The run timers will be reset.

Power Disruption

The GenePath system has a battery backed-up memory that retains the current program if the power is interrupted. If the program was in progress (not in PAUSE) when the power went down, the program will automatically resume after 2 minutes in PAUSE mode after power is restored.

Section 3 Sample Preparation

3.1 Agarose Blocks

Standard procedures for DNA preparation do not yield intact, high molecular weight DNA molecules. Large DNA molecules (chromosome-sized) are so fragile that they are sheared by mechanical forces during isolation. To prevent breakage of large DNA molecules, intact cells embedded in agarose are lysed and deproteinized *in situ*. The agarose matrix protects the embedded DNA from shear forces and provides an easy way to manipulate samples. Processed agarose plug-DNA inserts are loaded directly into sample wells of agarose electrophoresis gels.

3.2 Pathogen Group Reagent and Gel Kits

Bio-Rad has prepackaged reagent kit for the following pathogens:

Pathogen	Pathogen Group Reagent Kit
<i>Staphylococcus</i>	1
<i>Enterococcus</i>	1
<i>E. coli</i>	2
<i>Enterobacter</i>	3
<i>Pseudomonas</i>	3
<i>Candida</i>	4

The buffers, enzymes, and reagents needed to perform the analysis on these pathogens are provided in the Pathogen Group Reagent Kits (see the appropriate Pathogen Group Reagent Kit manual for information on sample preparation and running gels). Each reagent kit has sufficient material for 50 assays or 25 sample isolates.

Gel kits contain the agarose and buffers to pour and run samples using any of the Pathogen Group Reagent Kits. They are packaged separately to allow the maximum flexibility in run size. Each gel kit has sufficient material for 10 runs. The kit also contains ethidium bromide solution for staining the gels.

3.3 Casting the Gel

Casting the gel consists of using the following components: a casting stand with removable end plates; a platform on which the gel is cast, a comb and comb holder, and a frame which positions the gel and platform in the electrophoresis cell. The casting stand provided with the GenePath system is 14 cm (5.5") wide x 13 cm (5") long. Other sizes are available: 21 cm (8.5") wide x 14 cm (5.5") long, and 14 cm (5.5") wide x 21 cm (8.5") long.

The gel should be cast on a level surface. Bio-Rad's Leveling Table (catalog number 170-4046) is recommended for this purpose. For more detailed instructions, refer to the manual provided with the casting stand.

1. Slide the platform into the casting stand. There is no sidedness to the platform. Position one end gate over the screws protruding from the casting stand, insuring that the horizontal slot is facing toward the platform. Slide the edge of the platform into the slot, press down on the end gate, and gently tighten the screws.
2. Position the other end gate over the screws, and slide it toward the platform until the edge of the platform is inserted into the slot. Press down on the end gate, and gently tighten the screws. The slots force the platform against the bottom of the casting stand.
3. To attach the desired comb to the comb holder, place the comb over the two metal pins, and turn the screw clockwise. This causes one pin to move toward the screw, holding the comb in place. Adjust the height of the comb to 2 mm above the surface of the platform by loosening the screw (counterclockwise), then tightening when the comb is properly positioned. A thin plastic ruler makes a good height gauge.
4. Place the comb holder (with attached comb) into one of the two positioning slots on each side of the casting stand. Check that the bottom of the comb is at least 2 mm above the surface of the platform. Pour approximately 100 ml of the agarose solution (<60 °C) into the casting stand for a thickness of approximately 5-6 mm. Allow the gel to solidify for 30 minutes at room temperature.
5. Carefully remove the comb holder and comb; it is sometimes helpful to rock the holder back and forth slightly during its removal. Sample plugs can be added to the wells while the gel remains in the casting stand.

3.4 Hemocytometer Usage

In certain cases, it may be necessary to use a hemocytometer to obtain the appropriate cell concentration when the O.D. method (described in the Reagent Kit manuals) does not provide good results with a particular pathogen.

A hemocytometer is usually divided into nine large squares (Figure 3.1). Each large square is $1 \times 10^{-4} \text{ cm}^2$ or 0.1 mm^3 , one such square (A) is shown the figure with darkened borders. The large circle around the center square (B) represents your field of view at 100x power (10x objective lens, 10x eye piece). The center square (C) is subdivided into 25 smaller squares. The smaller circle in the center square represents your field of view at 400x power (40x objective lens, 10x eye piece). These 25 center squares are further divided into 16 squares.

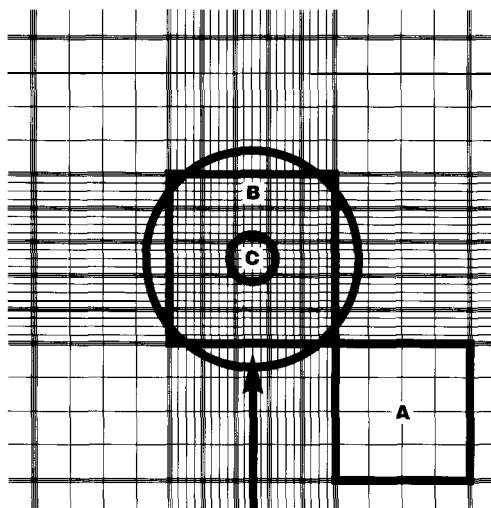


Fig. 3.1. Hemocytometer grid.

Bacteria and Yeast Cells:

Count five to ten of the 25 center squares, at 400x power, to get a representative sample of your cell suspension. You should have approximately 25 to 75 cells per square. The cells should be relatively free of clumps. Bacteria which naturally chain or grow in clusters are relatively easy to count and do not have to be dispersed by chemical or enzymatic methods. The Grams Crystal Violet aids in the visualization of bacteria.

Use the equations below to determine the cell concentration:

$$\frac{\text{Cells Counted}}{\text{Number of Squares}} = \text{Average Cells per Square}$$

$$\text{Average Cells per Square} \times 25 \text{ Squares} \times \text{Dilution Factor} \times 10^4 = \text{Cells per ml.}$$

$$\frac{\text{desired cell concentration}}{\text{actual cell concentration}} \times \text{ml. of plugs to be made} = \text{ml. of cell suspension to use.}$$

For example: 300 bacteria in 5 squares = average of 60 bacteria/square $\times$ 25 (squares) $\times$ 20 (dilution factor, yeast use 100 for dilution factor) $\times$ $10^4 = 3 \times 10^8$ bacteria per ml. So for 5 ml of plugs you need 5 ml $\times$ 5×10^7 cells final concentration $\div$ 3×10^8 actual cells concentration = 1.67 ml of cell suspension is required.

Section 4 Maintenance of Equipment

4.1 Replacing Electrodes

The gel chamber requires little maintenance except for rinsing after every run. Dirt and other build-up can be removed with laboratory detergent and a fine cloth. Care should be taken not to bend or break the electrodes. If one of the electrodes should break, or leak at the O-ring, it may be replaced. Additional electrodes (catalog number 170-3648) are available from Bio-Rad Laboratories.

To replace an electrode, turn the gel chamber upside down and remove the six screws. Lift off the base plate. Remove the hexagonal nut on the wire of the broken electrode, then remove the hexagonal nut on the electrode to be replaced. Push down firmly on the post to remove the old electrode. Turn the box over, insert the new electrode, pack with self-leveling silicone sealant (RTV-type silicone sealer available at most hardware stores), and replace the nut. Replace the wire and base plate.

If one of the pins to the serial cable bends, use tweezers to carefully straighten it. Replacement cables may be ordered from Bio-Rad.

4.2 Fuses

If the DC current during a run exceeds 500 mA entering the gel chamber, the 0.5 ampere FB (fast blow) fuse will blow, and error code F2 will be displayed. Replace the fuse by unscrewing the cartridge at the front of the GenePath system. Replace with 0.5 ampere FB fuse (two replacement fuses are provided). Make sure the GenePath system is off when replacing a fuse.

A power surge may cause the SB (slow blow) line fuse to blow. The LED lights on the GenePath system will not power on. The fuse is located at the rear of the drive module. Replace the fuse with a 3.15 A SB if your line voltage is 100 or 120 VAC, or a 1.6 A SB if your line voltage is 220 or 240 VAC.

Replacement fuses are available from a variety of sources, including Radio Shack® (Tandy Corporation). The 0.5 A FB fuse can be obtained as catalog number 270-1241, and the 3.15 A SB as catalog number 270-1173 from Radio Shack.

If the unit still does not operate, contact Bio-Rad Laboratories. Do not attempt to open and repair the GenePath system, or the warranty may be voided.

4.3 Maintenance of the Electrophoresis Cell

When the cell is not in use, even for short periods, all buffer should be removed to prevent damage to the plastics.

Section 5 Troubleshooting Guide

Equipment

Problem	Solution
No power	1. Check fuse at back of GenePath system 2. Check source of A/C power 3. Contact Bio-Rad Laboratories
No voltage across electrodes	1. Check that lid is on with AC power light on 2. Check that serial cable is fully inserted at both ends 3. Check that coiled interlock cable is firmly attached 4. Confirm that all leads are properly attached 5. Check HV fuse on front panel (0.5 A FB) and replace if necessary 6. Check that the unit is not set in pause mode
Gel floats away	1. Pump flow rate is too high. Adjust with variable speed pump.
No or low buffer flow	1. Look for kink in tubing 2. Check Cooling Module; buffer in the heat exchanger can freeze if the cooling module is cooling, but the pump is not on 3. Check pump connection and if the pump is on
Gel band patterns appear very distorted; lanes very curved,*	1. Foreign object in cell (remove thermometer, etc.) 2. Insufficient or non-uniform cooling due to low pump bands sharp, but slanting flow 3. Check that buffer is level with surface of gel. Use leveling feet 4. Replace damaged electrode 5. GenePath system fault; contact Bio-Rad Laboratories

* Slight distortion of the outermost lane is normal

Applications

Problem	Solution
Bands smeary or fuzzy	1. Improper program selected 2. Sample degraded. Impure enzymes, or wash cycles too short (agarose blocks); prepare new plugs. 3. Incomplete restriction digest; repeat with duplicate plugs 4. Sample overload. Adjust sample concentration.
Large DNAs not resolved	1. Improper program selected
High background in lanes	1. Insufficient washing of samples 2. Sample may be contaminated with RNA or other material; prepare new plugs 3. DNA concentration too high; adjust sample concentration
Distorted bands	1. Sample contains too high salt or detergent concentration; prepare new plugs 2. Buffer breakdown. Change after every run. 3. Wells were distorted. Recast gel. 4. Sample plugs were crushed when placed in wells 5. Pump flow rate too low. Check for kink along tubing.
Thick bands	1. Load less sample

Failure Codes

F-0	1. Error detected in the PS processor code during power up. 2. Contact Bio-Rad Laboratories
F-1	1. Error detected in the DAC processor code during power up. 2. Contact Bio-Rad Laboratories
F-2	1. The power supply control processor is not getting adequate feedback from the high voltage power supply during a run. Normal operation will continue and the display may be changed by pressing a display mode key. 2. Check fuses and safety interlock 3. Contact Bio-Rad Laboratories
F-3	1. The DAC external RAM has been corrupted (all parameters will reset to default values). Normal operation can be resumed by pressing any key on the right control panel. The battery RAM on the DAC board did not maintain its data when the power went down. If this occurs consistently, there may be a hardware problem on the board, or the battery may be bad. 2. Contact Bio-Rad Laboratories
F-4	1. A current overload condition has been detected. Check buffer concentration. If this occurs consistently, there may be a hardware problem. 2. Contact Bio-Rad Laboratories

Section 6

References

6.1 Applications in Pulsed Field Electrophoresis

The following are references in pulsed field electrophoresis, primarily from 1987-1989. The list surveys a wide area of applications and organisms, but is not exhaustive.

Organism	Reference numbers
<i>Acinetobacter</i>	99
<i>Aspergillus</i>	34
Bacteria	15, 70, 80, 116, 138, 142, 173, 186, 221, 226, 276, 283
<i>Bordetella</i>	135
<i>Candida albicans</i>	11, 23, 71, 141, 162, 163, 164, 178, 198, 223, 247, 281
<i>Campylobacter</i>	8, 219, 220, 299
<i>Caulobacter</i>	77
<i>Cryptococcus</i>	254,280
<i>Dictyostelium</i>	55
<i>Enterobacter</i>	105
<i>Enterococcus</i>	73, 109, 172, 182, 188, 195, 265, 293
<i>E. coli</i>	10, 24, 98, 183, 191, 303
<i>Giardia</i>	2, 127, 140, 222
<i>Histoplasma</i>	255
<i>Legionella</i>	227
<i>Leishmania</i>	3, 87, 97, 101, 152, 197, 231, 239
<i>Leptospira</i>	113
<i>Listeria</i>	32, 33, 35, 42, 121
<i>Moraxella</i>	213
Mouse	30, 31, 47, 122, 137, 146,260
<i>Mycobacterium</i>	37,110, 172, 203, 241, 302
<i>Mycoplasma</i>	13, 43, 166,205
<i>Neurospora</i>	14.5, 190
<i>Nocardia</i>	256
<i>Plasmodium</i>	17, 61, 130, 204, 237, 248
<i>P. carinii</i>	119,120
<i>Pseudomonas</i>	8, 14, 26, 27, 103, 187, 199, 215, 295
<i>S. cerevisiae</i>	4, 39, 49, 79, 131, 139, 177, 179, 225, 238, 243, 266, 282, 300, 301
<i>Shigella</i>	67
<i>Staphylococcus</i>	41, 67, 76, 95, 124, 125, 133, 155, 157, 181,203, 224, 236, 259, 261, 288, 290, 291
<i>S. pombe</i>	46, 78, 106, 108, 171, 246, 270
<i>Tetrahymena</i>	57
<i>Torulopsis</i>	136
<i>Trypanosoma</i>	165, 278
<i>Ureaplasma</i>	53

Application	Reference Numbers
CHEF	21, 34, 44, 46, 48, 49, 55, 57, 58, 61, 64, 82, 83, 84, 87, 129, 139, 142, 143, 145, 158, 174, 177, 200, 205, 210, 214, 225, 231, 238, 239, 245, 248, 250, 285, 286, 292, 294
Mapping	1, 30, 85, 92, 131, 196, 206, 287
Minute chromosomes	44,165
OFAGE	39, 77, 79, 115, 118, 131,217
On-off pulsing	137,263
PACE	18, 50, 147
PFGE	28, 74,151, 201, 204, 206, 228, 230, 275, 270, 294, 123
PFG in acrylamide	128
PHOGE	12
Preparative	64
Ramps	29, 158,209
Restriction enzymes	154, 173
Review of PFGE	148
RFGE	4, 7, 16, 175, 177, 232, 234, 262
RFLP polymorphism	14, 22, 59, 62, 74, 91, 151, 161,223
Sample preparation - bacteria	84, 244, 249, 283
Sample preparation - cell lines	244
Sample preparation - general	214, 230, 244, 274
Sample preparation - tissues,	156,244
Sample preparation-YACs	64
Secondary structure analysis	48, 170, 186
Single strand DNA	263,268
Size standards	58, 107, 168,289
Strain characterization	54, 97, 102, 174
TAFE	75, 88, 89, 193, 218, 260
Theory	25, 38, 63, 69, 111, 117, 149, 150, 169, 184, 189, 190, 232, 252, 253, 257, 258, 271, 284, 6, 5
Two-D PFG	17,297
virus	25, 214, 132, 51
Visualization of DNA (microscope)	229,245
Yeast artificial chromosomes (YACs)	17, 28, 60, 64, 142, 143, 144,158, 167, 212, 267, 286

6.2 Reference List for Pulsed Field Electrophoresis

1. Abel, K. J. and Gross, K. 'W., *Nucl. Acids Res.*, **16**, 2111-2126 (1988).
2. Adams, R. A., Nash, T. E. and Wellem, T. E., *Nucl. Acids Res.*, **16**, 4555-4565 (1988).
3. Aebischer, T., Moody, S. F. and Handman, E., *Infect. Immun.*, **61**, 220-226 (1993).
4. Albig, W. and Entian, K-D., *Gene*, **73**, 141-152 (1988).
5. Allardet-Servent, A., Bouziges, N., Carles-Nut-It, M. J., Bourg, G., Gouby, A. and Ramuz, M., *J. Clin. Microbiol.*, **27**, 2057-2061 (1989).
6. Allardet-Set-vent, A., Bourg, G., Ramuz, M., Pages, M., Bellis, M. and Roizes, G., *J. Bacteriol.*, **170**, 4603-4607 (1988).
7. Amler, L. C. and Schwab, M., *Molec. and Cell. Biol.*, **9**, 4903-4913 (1989).
8. Alvarado, C. J., Stolz, S. M. and Maki D. G., *Am. J. Med.*, **91**, 272S-280S (1991).
9. Anderson, D. J., Kuhns, J. S., Vasil, M. L., Gerding, D. N. and Janoff, E. N., *J. Clin. Microbiol.*, **29**, 648-649 (1991).
10. Arbeit, R. D., Arthur, M., Dunn, R., Kim, C., Selander, R. K. and Goldstein, R., *J. Infect. Dis.*, **161**, 230-235 (1990).
11. Asakura, K., Iwaguchi, S., Homma, M., Sukai, T., Higashide, K. and Tanaka, K., *J. Gen. Microbiol.*, **137**, 2531-2538 (1991).
12. Bancroft 1. and Wolk, C. P., *Nucl. Acids Res.*, **16**, 7405-7417 (1988).
13. Bautsch, W., *Nucl. Acids Res.*, **16**, 11461-11467 (1988).
14. Bautsch, W., Grothues, D. and Tummler, B., *FEMS Microbiol. Let.*, **52**, 255-258 (1988).
15. Beverley, S., *Anal. Biochem.*, **177**, 110-114 (1989).
16. Bickmore, W. A., Maule, J. C., Van Heyningen, V. and Porteus, D. J., *Somatic Cell. Mol. Genet.*, **15**, 229-236 (1989).
17. Biggs, B. A., Kemp, D. J. and Brown, G. V., *Proc. Natl. Acad. Sci. USA*, **86**, 2428-2432 (1989).
18. Birren, B. W., Hood, L. and Lai, E., *Electrophoresis*, **10**, 302-309 (1989).
19. Birren, B. W., Lai, E., Hood, L. and Simon, M. I., *Anal. Biochem.*, **177**, 282-286 (1989).
20. Bleumink-Pluym, N., ter Laak, E. A. and van der Zeijst, B. A., *J. Clin. Microbiol.*, **28**, 2012-2016 (1990).
21. Blocher, D., Einspenner, M. and Zajackowski, J., *Int. J. Radiat. Biol.*, **56**, 437-448 (1989).
22. Blonden, L. A., Dunnen, J. T., van Paasen, H. M., Wapenaar, M. C., Grootsholten, P. M., Ginjaar, H. B., Bakker, E., Pearson, P. L. and van Ommen, G. J., *Nucl. Acids Res.*, **17**, 5611-5621 (1989).
23. Blumberg, H. M., Hendershot, E. F. and Lott, T. J., *Diagn. Microbiol. Infect. Dis.*, **15**, 545-547 (1992).
24. Bohm, H. and Karch, H., *J. Clin. Microbiol.*, **30**, 2169-2172 (1992).
25. Bostock, C. J., *Nucl. Acids Res.*, **16**, 4239-4252 (1988).
26. Boukadida, J., De Montalembert, M., Lenoir, G., Scheinmann, P., Veron, M. and Elerche, P., *J. Med. Microbiol.*, **38**, 29-33 (1993).
27. Boukadida, J., de Montalembert, M., Gaillard, J. L., Gobin, J., Grimont, F., Girault, D., Veron, M. and Berche, P., *J. Clin. Microbiol.*, **29**, 2068-2071 (1991).
28. Bowcock, A. M., Herbert, J. M., Wijsman, E., Gadi, I., Cavalli-Sforza, L. L. and Boyd, C. D., *Proc. Natl. Acad. Sci. USA*, **85**, 2701-2705 (1988).
29. Bray, P. F., Barsh, G., Rosa, J.-P., Luo, X. Y., Magenis, E. and Shuman, M. A., *Proc. Natl. Acad. Sci. USA*, **85**, 8683-8687 (1988).
30. Brockdorf, N., Amar, L. C. and Brown, S. D., *Nucl. Acids Res.*, **17**, 1315-1326 (1989).
31. Brody, H. and Carbon, C., *Proc. Natl. Acad. Sci. USA*, **86**, 6260-6263 (1989).
32. Brosch, R., Buchrieser, C., Sixl-Voigt, B. and Rocourt, J., *Int. J. Med. Microbiol.*, **275**, 557-560 (1991).
33. Brosch, R., Buchrieser, C. and Rocourt, J., *Res. Microbiol.*, **142**, 667-675 (1991).
34. Brown, W. R. and Bird, A. P., *Nature*, **322**, 477-481 (1986).
35. Buchrieser, C., Brosch, R. and Rocourt, J., *Int. J. Food Microbiol.*, **14**, 297-304 (1991).
36. Burke, D. T., Carle, G. F. and Olson, M. V., *Science*, **236**, 806-812 (1987).

37. Bums, D. N., Wallace, Jr., R. J., Schultz, M. E., Zhang, Y. S., Zubairi, S. Q., Pang, Y. J., Gibert, C. L., Brown, B. A., Noel, E. S. and Gordin, F. M., *Am. Rev. Respir. Dis.*, 144, 1153-1159 (1991).
38. Cantor, C. R., Gaal, A. and Smith, C. L., *Biochem.*, **27**, 9216-9221 (1988).
39. Carle, G. F. and Olson, M. V., *Nucl. Acids Res.*, **12**, 5647-5664 (1984).
40. Carle, G. F. and Olson, M. V., *Science*, **232**, 65-68 (1986).
41. Carles-Nurit, M. J., Christophle, B., Broche, S., Gouby, A., Bouziges, N. and Ramuz, M., *J. Clin. Microbiol.*, **30**, 2092-2096 (1992).
42. Carriere, C., Allardet-Servent, A., Bourg, G., Audurier, A. and Ramuz, M., *J. Clin. Microbiol.*, **29**, 1351-1355 (1991).
43. Chen, T. L. and Manuelidlis, L., *Genomics*, **4**, 430-433 (1989).
44. Chen, X. and Finch, L. R., *J. Bacteriol.*, **171**, 2876-2878 (1989).
45. Cheng, J.-F., Smith, C. L. and Cantor, C. R., *Nucl. Acids Res.*, **17**, 6109-6127 (1989).
46. Chikashige, Y., Kinoshita, N., Nakaseko, Y., Matsumoto, T., Murakami, S., Niwa, O. and Yanagida, M., *Cell*, **57**, 739-752 (1989).
47. Chou, H. S., Nelson, C. A., Godambe, S. A., Chaplin, D. D. and Loh, D. Y., *Science*, **238**, 545-547 (1987).
48. Chu, G., *Electrophoresis*, **10**, 290-295 (1989).
49. Chu, G., Vollrath, D. and Davis, R., *Science*, **234**, 1582-1585 (1986).
50. Clark, S. M., Lai, E., Birren, B. W., Hood, L. and Simon, M. I., *Science*, **241**, 1203-1205 (1988).
51. Clarke, P., Rabkin, S. D., Inman, M. V., Mahalingam, R., Cohrs, R., Wellish, M. and Gilden, D. H., *Virology*, **190**, 597-605 (1992).
52. Clevers, H. C., Dunlap, S., Wileman, T. E. and Terhorst, C., *Proc. Natl. Acad. Sci. USA*, **85**, 8156-8160 (1988).
53. Cocks, B. G., Pyle, L. E. and Finch, L. R., *Nucl. Acids Res.*, **17**, 6713-6719 (1989).
54. Coetzee, D. J., Kock, J. L. and Pretorius, G. H., *J. Microbiol. Methods*, **7**, 219-225 (1987).
55. Cole, R. A. and Williams, K. L., *Nucl. Acids Res.*, **16**, 4891-4902 (1988).
56. Compton, D. A., Weil, M. M., Jones, C., Riccardi, V. M., Strong, L. C. and Saunders, G. F., *Cell*, **55**, 827-836 (1988).
57. Conover, R. K. and Brunk, C. F., *Molec. and Cell. Biol.*, **6**, 900-905 (1986).
58. Cooney, C. A., Galbraith, J. L. and Bradbury, M. E., *Nucl. Acids Res.*, **17**, 5412 (1989).
59. Corcoran, L., Forsyth, K., Bianco, A., Brown, G. and Kemp, D., *Cell*, **44**, 87-95 (1986).
60. Coulson, A., Waterson, R., Kiff, J., Sulston, J. and Kohara, Y., *Nature*, **335**, 184-186 (1988).
61. Cowman, A. F., Morry, M. J., Biggs, B. A., Cross, G. A. and Foote, S. J., *Proc. Natl. Acad. Sci. USA*, **85**, 9109-9113 (1988).
62. Craig, J., Fowler, S., Skinner, J. D., Burgoyne, L. A. and McInnes, J. L., *Applied and Theor. Electro.*, **1**, 23-28 (1988).
63. Crater, G. D., Gregg, M. C. and Holzworth, G., *Electrophoresis*, **10**, 310-314 (1989).
64. Cuoto, L. B., Spangler, E. A. and Rubin, E. M., *Nucl. Acids Res.*, **17**, 8010 (1989).
65. De Jonckheere, J. F., *Parasitol. Res.*, **76**, 55-62 (1989).
66. Deaven, L. L., Hildebrand, C. E., Longmire, J. L. and Moyzis, R. K., *Abstract # 77: Human Genome I Conference*, San Diego (1989).
67. Deighton, M., Pearson, S., Capstick, J., Spelman, D. and Borland, R., *J. Clin. Microbiol.*, **30**, 2385-2390 (1992).
68. Denko, N., Giaccia, A., Peters, B. and Stamato, T. D., *Anal. Biochem.*, **178**, 172-176 (1989).
69. Deutsch, J. M., *Science*, **240**, 922-924 (1988).
70. Dingwall, A. and Shapiro, L., *Proc. Natl. Acad. Sci. USA*, **86**, 119-123 (1989).
71. Doebbeling, B. N., Lehmann, P. F., Hollis, R. J., Wu, L. C., Widmer, A. F., Voss, A. and Pfaller, M. A., *Clin. Infect. Dis.*, **16**, 377-383 (1993).
72. Doggett, N. A., Smith, C. L. and Cantor, C. R., *Nucl. Acids Res.*, **20**, 859-864 (1992).
73. Donabedian, S. M., Chow, J. W., Boyce, J. M., McCabe, R. E., Markowitz, S. M., Coudron, P. E., Kuritz, A., Pierson, C. L. and Zervos, M. J., *J. Clin. Microbiol.*, **30**, 2757-2761 (1992).

74. Dunham, I., Sargent, C. A., Trowsdale, J. and Cambell, R. D., *Proc. Natl. Acad. Sci. USA*, **84**, 7237-7241 (1987).
75. Edman, J. C., Edman, U., Cao, M., Lundgren, B., Kovacs, J. A. and Santi, D. V., *Proc. Natl. Acad. Sci. USA*, **86**, 8625-8629 (1989).
76. el-Adhami, W., Roberts, L., Vickery, A., Inglis, B., Gibbs, A. and Stewart P. R., *J. Gen. Microbiol.*, **137**, 2713-2720 (1991).
77. Ely, B. and Gerardot, C. J., *Gene*, **68**, 323-333 (1988).
78. Fan, J.-B., Chikashige, Y., Smith, C. L., Niwa, O., Yanagida, M. and Cantor, C. R., *Nucl. Acids Res.*, **17**, 2801-2818 (1988).
79. Fasullo, M. and Davis, F. W., *Molec. and Cell. Biol.*, **8**, 4370-4380 (1988).
80. Ferdows, M. S. and Barbour, A. G., *Proc. Natl. Acad. Sci. USA*, **86**, 5969-5973 (1989).
81. Ferrin, L. J. and Camerini-Otero, D., *Science*, **254**, 1494-1497 (1991).
82. Ferris, S., Freeby, S., Zoller, P., Ragsdale, C. and Stevens, A., *Amer. Biotech. Lab.*, **7**, 36-42 (1989).
83. Ferris, S., Sparrow, L. and Stevens, A., *Australian J. Biotechnol.*, **3**, 33-35 (1989).
84. Flanagan, J. L., Ventra, L. and Weiss, A. S., *Nucl. Acids Res.*, **17**, 814 (1989).
85. Fountain, J. W., Wallace, M. R., Bruce, M. A., Seizinger, B. R., Menon, A. G., Gusella, J. F., Michels, V. V., Schmidt, M. A., Dewald, G. W. and Collins, F. S., *Science*, **244**, 1085-1087 (1989).
86. Fulton, T. R., Bowcock, A. M., Smith, D. R., Daneshvar, L., Green, P., Cavalli-Sforza, L. L. and Donis-Keller, H., *Nucl. Acids Res.*, **17**, 271-284 (1989).
87. Galindo, O., Mons, B. and Van Der Berg, F. M., *Exp. Parasitol.*, **34**, 245-252 (1989).
88. Gardiner, K., Laas, W. and Patterson, D., *Somat. Cell. and Molec. Gen.*, **12**, 185-195 (1986).
89. Cardiner, K. and Patterson, D., *Electrophoresis*, **10**, 296-301 (1989).
90. Garza, D., Ajioka, J. W., Burke, D. T. and Hartl, D. L., *Science*, **246**, 641-646 (1989).
91. Gejman, P. V., Sitaram, N., Hsieh, W.-T., Gelemter, J. and Gershon, E. S., *Applied and Theoret. Electrophor.*, **1**, 29-34 (1988).
92. Gemmill, R. M., Coyle-Morris, J. F., McPeck, F. D., Ware-Urbe, L. F. and Hecht, F., *Gene Anal. Techn.*, **4**, 119-131 (1987).
93. Gessler, M., Simola, K. O. and Bruns, G. A., *Science*, **244**, 1575-1577 (1989).
94. Gilley, D., Preer, J. R., Aufderheide, K. J. and Polisky, B., *Molec. and Cell. Biol.*, **8**, 4765-4772 (1988).
95. Goering, R. V. and Winters, M. A., *J. Clin. Microbiol.*, **30**, 577-580 (1992).
96. Goering, R. V. and Duensing, T. D., *J. Clin. Microbiol.*, **28**, 426-429 (1990).
97. Gomez-Eichelmann, M. C., Holz, G., Beach, D., Simpson, A. M. and Simpson, L., *Molec. and Biochem. Parasitol.*, **27**, 143-158 (1988).
98. Gordillo, M. E., Reeve, G. R., Pappas, J., Mathewson, J. J., DuPont, H. L. and Murray B. E., *J. Clin. Microbiol.*, **30**, 889-893 (1992).
99. Gouby, A., Carles-Nurit, M. J., Bouziges, N., Bourg, G., Mesnard, R., Bouvet, P.J., *J. Clin. Microbiol.*, **30**, 1588-1591 (1992).
100. Graham, M. Y., Otani, T., Boime, I., Olsen, M. V., Carle, G. F. and Chaplin, D., *Nucl. Acids Res.*, **15**, 4437-4448 (1987).
101. Groggl, M., Kreutzer, R. D., McHugh, C. P., Martin, R. K., *Am. J. Trop. Med. Hyg.*, **45**, 714-722 (1991).
102. Grothues, D. and Tummler, B., *FEMS Microbiol. Lett.*, **48**, 419-422 (1987).
103. Grothues, D., Koopmann, U., von der Hardt, H. and Tummler, B., *J. Clin. Microbiol.*, **26**, 1973-1977 (1988).
104. Guzman, P. and Ecker, J. R., *Nucl. Acids Res.*, **16**, 11091-11105 (1988).
105. Haertl, R. and Bandlow, G., *J. Clin. Microbiol.*, **31**, 128-133 (1993).
106. Hahnenberger, K. M., Baum, M. P., Polizzi, C. M., Carbon, J. and Clarke, L., *Proc. Natl. Acad. Sci. USA*, **86**, 577-581 (1989).
107. Hanlon, D. J., Smardon, A. M. and Lane, M. J., *Nucl. Acids Res.*, **17**, 5413 (1989).
108. Harris, A. and Bentley, D. R., *Nucl. Acids Res.*, **16**, 4172 (1988).

109. Hayden, M. K., Trenholme, G. M., Schultz, J. E. and Sahm, D. F., *J. Infect. Dis.*, 167, 1224-1227 (1993).
110. Hector, J. S., Pang, Y., Mazurek, G. H., Zhang, Y., Brown, B. A. and Wallace, Jr., R. J., *J. Clin. Microbiol.*, 30, 1250-1255 (1992).
111. Heller, C. and Pohl, F. M., *Nucl. Acids Res.*, 17, 5989-6003 (1989).
112. Hennekes, H. and Kuhn, S., *Anal. Biochem.*, 183, 80-83 (1989).
113. Herrmann, J. L., Bellenger, E., Perolat, P., Baranton, G. and Saint Girons, I., *J. Clin. Microbiol.*, 30, 1696-1702 (1992).
114. Hightower, R., Metge, D. W. and Santi, D. V., *Nucl. Acids Res.*, 15, 8387-8398 (1987).
115. Hightower, R. and Santi, D.V., *Electrophoresis*, 10, 283-289 (1989).
116. Hockett, R. D., de Villartay, J-P., Pollock, K., Poplack, D. G., Cohen, D. I. and Korsmeyer, S. J., *Proc. Natl. Acad. Sci. USA*, 85, 9694-9698 (1988).
117. Hofman, M. D., Schalkwyk, L. C. and Doolittle, W. F., *Nucl. Acids Res.*, 14, 6983-7000 (1986).
118. Holzwarth, G., McKee, C., Steiger, S. and Crater, G., *Nucl. Acids Res.*, 15, 10031-10044 (1987).
119. Hong, S. T., Ryu, J. S., Chai, J. Y. and Lee S. H., *Kisaengchunghak Chapchi*, 30, 283-288 (1992).
120. Hong, S. T., Steele, P. E., Cushion, M. T., Walzer, P. D., Stringer, S. L. and Stringer, I. R., *J. Clin. Microbiol.*, 28, 1785-1795 (1990).
121. Howard, P. J., Harsono, K. D. and Luchansky, J. B., *Appl. Environ. Microbiol.*, 58, 709-712 (1992).
122. Ichikawa, H., Shimizu, K., Saito, A., Wang, D., Oliva, R., Kobayashi, H., Kaneko, Y., Miyoshi, H., Smith, C. L., Cantor, C. R. and Ohki, M., *Proc. Natl. Acad. Sci. USA*, 89, 23-27 (1992).
123. Ichiyama, S. and Ohta M., *Nippon Rinsho*, 50 Suppl, 143-148 (1992).
124. Ichiyama, S., *Nippon Rinsho*, 50, 992-997 (1992).
125. Ichiyama, S., Ohta, M., Shimokata, K., Kato, N. and Takeuchi J., *J. Clin. Microbiol.*, 29, 2690-2695 (1991).
126. Inoko, H. and Trowsdale, J., *Nucl. Acids Res.*, 21, 8957-8963 (1987).
127. Isaac-Renton, J. L., Cordeiro, C., Sarafis, K. and Shahriari, H., *J. Infect. Dis.*, 167, 431-440, 1993.
128. Ito, T. and Sakaki, Y., *Nucl. Acids Res.*, 16, 9177-9184 (1988).
129. Jabs, E. W., Goble, C. A. and Cutting, G. R., *Proc. Natl. Acad. Sci. USA*, 86, 202-206 (1989).
130. Janse, C. J., Boorsma, E. G., Ramesar, J., Van Vianen, P., Van der Meer, R., Zenobi, P., Casaglia, O., Mons, B. and Van der berg, F. M., *Exp. Purusitol.*, 68, 274-282 (1989).
131. Kaback, D. B., Steensma, H. Y. and DeJonge, P., *Proc. Natl. Acad. Sci. USA*, 86, 3694-3698 (1989).
132. Kane, E. M. and Shuman, S., Program in Molecular Biology, *J. Virol.*, 66, 5752-5762 (1992).
133. Kaufhold, A., Livdahl, C. and Ferrieri, P., *Int. J. Med. Microbiol. Virol. Purusitol. Infect. Dis.*, 277, 309-319 (1992).
134. Kenrick, S., Patterson, M., Speer, A., Fischbeck, K. and Davies, K., *Cell*, 48, 351-375 (1987).
135. Khattak, M. N., Matthews, R. C. and Bumie, J. P., *BMJ*, 304, 813-815 (1992).
136. Khattak, M. N., Bumie, J. P., Matthews, R. C. and Oppenheim, B. A., *J. Clin. Microbiol.*, 30, 2211-2215 (1992).
137. Kingsmore, S. F., Snoddy, J., Choubey, D., Lengyel, P. and Seldin, M. F., *Immunogenetics*, 30, 169-174 (1989).
138. Kohara, Y., Akiyama, K. and Isono, K., *Cell*, 50, 495-508 (1987).
139. Kolakowski, L. F., Schloesser, M. and Cooperman, B. S., *Nucl. Acids Res.*, 16, 10441-10452 (1988).
140. Korman, S. H., Le Blancq, S. M., Deckelbaum, R. J. and Van der Ploeg, L. H., *J. Clin. Invest.*, 89, 1725-1733 (1992).
141. Kurtz, M., Cortelyou, M., Miller, S., Lai, M. and Kirsch, D., *Molec. and Cell. Biol.*, 7, 209-217 (1987).
142. Kuspa, A., Vollrath, D., Cheng, Y. and Kaiser, D., *Proc. Natl. Acad. Sci. USA*, 86, 8917-8921 (1989).
143. Labella, T. and Schlessinger, D., *Genomics*, 5, 752-760 (1989).
144. Lai, E., *Nucl. Acids Res.*, 17, 8008 (1989).
145. Lai, E., *Nucl. Acids Res.*, 18, in press (1990).
146. Lai, E., Barth, R. K. and Hood, L., *Proc. Natl. Acad. Sci. USA*, 84, 3846-3850 (1987).

147. Lai, E., Birren, B. W., Clark, S. M. and Hood, L., *Nucl. Acids Res.*, **16**, 10376 (1988).
148. Lai, E., Birren, B. W., Clark, S. M., Simon, M. I. and Hood, L., *BioTechniques*, **7**, 34-42 (1989).
149. Lalande, M., Noolandi, J., Turmel, C., Brousseau, R., Rousseau, J. and Slater, G. W., *Nucl. Acids Res.*, **16**, 5427-5437 (1988).
150. Lalande, M., Noolandi, J., Turmel, C., Rousseau, J. and Slater, G. W., *Proc. Natl. Acad. Sci. USA*, **84**, 8011-8015 (1987).
151. Ledbetter, D. H., Ledbetter, S., van Tuinen, P., Summers, K. M., Robinson, T. J., Nakamura, Y., Wolff, R., White, R., Barker, D. F., Wallace, M., Collins, F. S. and Dobyns, W. B., *Proc. Natl. Acad. Sci. USA*, **86**, 5136-5140 (1989).
152. Lee, M. G.-S., Atkinson, B. L., Giannini, H. and Van der Ploeg, L. H., *Nucl. Acids Res.*, **16**, 9567-9586 (1988).
153. Levene, S. and Zimm, B., *Proc. Natl. Acad. Sci. USA*, **84**, 4054-4057 (1987).
154. Levine, J. D. and Cech, C. L., *Biotechnology*, **7**, 1033-1036 (1989).
155. Lina, B., Vandenesch, F., Etienne, J., Kreiswirth, B. and Fleurette, J., *FEMS Microbiol. Lett*, **71**, 133-138 (1992).
156. Lindsten, T., Lee, N. E. and Davis, M. M., *Proc. Natl. Acad. Sci. USA*, **84**, 7639-7643 (1987).
157. Linhardt, F., Ziebuhr, W., Meyer, P., Witte, W. and Hacker, J., *FEMS Microbiol. Lett*, **74**, 181-185 (1992).
158. Little, R. D., Porta, G., Carle, G. F., Schlessinger, D. and Urso, M., *Proc. Natl. Acad. Sci.*, **86**, 1598-1602 (1989).
159. Looney, J. E., Chi, M., Leu, T.-H., Flintoff, W. F., Troutman, W. B. and Hamlin, J. L., *Molec. and Cell. Biol.*, **8**, 5268-5279 (1988).
160. Lupski, J. R., Montes de Oca-Luna, R., Slaugenhaupt, S., Pentao, L., Guzzetta, V., Trask, B. J., Saucedo-Cardenas, O., Barker, D. F., Killian, J. M., Garcia, C. A., Chakravarti, A., and Patel, P. I., *Cell*, **66**, 219-232 (1991).
161. Maeda, N., McEvoy, S. M., Harris, H. F., Huisman, T. H. and Smithies, O., *Proc. Natl. Acad. Sci. USA*, **83**, 7359-7399 (1986).
162. Magee, B. B., Koltin, Y., Gorman, J. A. and Magee, P. T., *Molec. and Cell. Biol.*, **8**, 4721-4726 (1988).
163. Magee, B. B. and Magee, P. T., *J. Gen. Microbiol.*, **133**, 425-430 (1987).
164. Magee, P. T., Rikkerink, E. H. and Magee, B. B., *Anal. Biochem.*, **175**, 361-372 (1988).
165. Majiwa, P. A., Young, J. R., Hamers, R. and Mattysens, G., *Gene*, **41**, 183-192 (1986).
166. Maniloff, J., *Nucl. Acids Res.*, **17**, 1268 (1989).
167. Marchuk, D. and Collins, F. S., *Nucl. Acids Res.*, **16**, 7743 (1988).
168. Mathew, K. M., Hui, C.-F., Smith, C. L. and Cantor, C. R., *Biochem.*, **27**, 9222-9226 (1988).
169. Mathew, M. K., Smith, C. L. and Cantor, C. R., *Biochem.*, **27**, 9204-9210 (1988).
170. Mathew, M. K., Smith, C. L. and Cantor, C. R., *Biochem.*, **27**, 9210-9216 (1988).
171. Matsumoto, T., Fukui, K., Niwa, O., Sugawara, N., Szostak, J. W. and Yanagida, M., *Molec. and Cell. Biol.*, **7**, 4424-4430 (1987).
172. Mazurek, G. H., Hartman, S., Zhang, Y., Brown, B. A., Hector, J. S., Murphy, D. and Wallace, Jr., R. J., *J. Clin. Microbiol.*, **31**, 390-394 (1993).
173. McClelland, M., Jones, R., Patel, Y. and Nelson, M., *Nucl. Acids Res.*, **15**, 5985-6005 (1987).
174. McCluskey, K. and Mills, D., *J. Cell Biochem. Suppl.*, **O**, **13**, Part E (1989).
175. Miles, J. S., Bickmore, W., Brook, J. D., McLaren, A. W., Meeham, R. and Wolf, C. R., *Nucl. Acids Res.*, **17**, 2907-2917 (1989).
176. Miranda, A. G., Singh, K. V. and Murray, B. E., *J. Clin. Microbiol.*, **29**, 2752-2757 (1991).
177. Monia, B. P., Haskell, K. M., Ecker, J. R., Ecker, D. J. and Crooke, S. T., *Nucl. Acids Res.*, **17**, 3611 (1989).
178. Monod, M., Porchet, S. and Baudraz-Rosselet, F., *J. Med. Microbiol.*, **32**, 123-129 (1990).
179. Mortimer, R. K. and Schild, D., *Microbiol. Rev.*, **49**, 181-213 (1985).
180. Muller, U., Stephan, D., Philippsen, P. and Steinmetz, M., *EMBO J.*, **6**, 369-373 (1987).

181. Mulligan, M. E. and Arbeit, R. D., *Infect. Control. Hosp. Epidemiol.*, **12**, 20-28 (1991).
182. Murray, B. E., Singh, K. V., Markowitz, S. M., Lopardo, H. A., Patterson, J. E., Zervos, M. J., Rubeglio, E., Eliopoulos, G. M., Rice, L. B., Goldstein, F. W., *et. al.*, *J. Infect. Dis.*, **163**, 780-785 (1991).
183. Nada, T., Ichiyama, S., Iida, E., Kadoya, M., Takeuchi, J. and Ohta, M., *Kansenshogaku Zasshi*, **66**, 465-469 (1992).
184. Noolandi, J., Slater, G. W., Lim, H. A. and Viovy, J. L., *Science*, **243**, 1456-1458 (1989).
185. O'Connell, P., Leach, R., Cawthon, R. M., Culver, M., Stevens, J., Viskochil, D., Fournier, R. K., Rich, D. C., Ledbetter, D. H. and White, R., *Science*, **244**, 1085-1087 (1989).
186. Ohki, M. and Smith, C. L., *Nucl. Acids Res.*, **17**, 3479-3490 (1989).
187. Ojeniyi, B., Hoiby, N. and Rosdahl, V. T., *APMIS*, **99**, 492-498 (1991).
188. Okahashi, N., Sasakawa, C., Okada, N., Yamada, M., Yoshikawa, M., Tokuda, M., Takahashi, I. and Koga, T. *J. Gen. Microbiol.*, **136**, 22 17-2223 (1990).
189. Olschwang, S. and Thomas, G., *Nucl. Acids Res.*, **17**, 2363 (1989).
190. Orbach, M., Vollrath, D., Davis, R. and Yanofsky, C., *Molec. and Cell. Biol.*, **8**, 1469-1473 (1988).
191. Ott, M., Bender, L., Blum, G., Schmittroth, M., Achtman, M., Tschape, H. and Hacker, J., *Infect. Immun.*, **59**, 2664-2272 (1991).
192. Patarapotikul, J. and Langsley, G., *Nucl. Acids Res.*, **16**, 4331-4340 (1988).
193. Patterson, D., Gardiner, K., Kao, F.-Z.T., Tanzi, R. and Watkins, P., *Proc. Natl. Acad. Sci. USA*, **85**, 8266-8270 (1988).
194. Patterson, M., Schwartz, C., Bell, M., Sauer, S., Hofker, M., Trask, B., van den .Engh, G. and Davies, K. E., *Genomics*, **1**, 297-306 (1987).
195. Patterson, J. E., Singh, K. V. and Murray, B. E., *J. Clin. Microbiol.*, **29**, 2513-2516 (1991).
196. Peltz, G. A., Grundy, H. O., Lebo, R. V., Yssel, H., Barsch, G. S. and Moore, K. W., *Proc. Natl. Acad. Sci. USA*, **86**, 1013-1017 (1989).
197. Petrillo-Peixoto, M. L. and Beverley, S. M., *Molec. and Cell. Biol.*, **8**, 5188-5199 (1988).
198. Pittet, D., Monod, M., Filthuth, I., Frenk, E., Suter, P.M. and Auckenthaler, R., *Am. J. Med.*, **91**, 256S-263S (1991).
199. Poh, C. L., Yeo, C. C. and Tay, L., *Eur. J. Clin. Microbiol. Infect. Dis.*, **11**, 817-822 (1992).
200. Pohl, T. M., Zimmer, M., MacDonald, M. E., Smith, B., Bucan, M., Poustka, A., Volinia, S., Zehetner, G., Wasmuth, J. J., Gusella, J., Lehrarch, H. and Frischauf, A.-M., *Nucl. Acids Res.*, **16**, 9185-9198 (1988).
201. Pologe, L. G. and Ravetch, J. V., *Cell*, **55**, 869-874 (1988).
202. Prevost, G., Jaulhac, B. and Piemont, Y., *J. Clin. Microbiol.*, **30**, 967-973 (1992).
203. Prevost, G., Pottecher, B., Dahlet, M., Bientz, M., Mantz, J. M. and Piemont, Y., *J. Hosp. Infect.*, **17**, 255-269 (1991).
204. Pritchard, C. A., Goodfellow, P. J. and Goodfellow, P. N., *Nature*, **238**, 273-275 (1987).
205. Pyle, L. E., Corcoran, L. N., Cocks, B. G., Bergemann, A. D., Whitley, J. C. and Finch, L. R., *Nucl. Acids Res.*, **16**, 6015-6025 (1988).
206. Rappold, G. A. and Lehrach, H., *Nucl. Acids Res.*, **16**, 5361-5377 (1988).
207. Reschke, D. K., Frazier, M. E., and Mallavia, L. P., *Acta Virol.*, **35**, 519-525 (1991).
208. Resnick, M. A., Skaanild, M. and Nilsson-Tillgren, T., *Proc. Natl. Acad. Sci. USA*, **86**, 2276-2280 (1989).
209. Richards, J. E., Gilliam, T. C., Cole, J. L., Drumm, M. L., Wasmuth, J. J., Gusella, J. I. and Collins, F. S., *Proc. Natl. Acad. Sci. USA*, **85**, 6437-6441 (1988).
210. Richmond, T., *Biotechnol. Lab. News*, Feb. 14 (1989).
211. Ridley, R. G., White, J. H., McAleese, S. M., Goman, M., Alano, P., de Vries, E., and Kilbey, B. J., *Nucl. Acids Res.*, **19**, 6731-6736 (1991).
212. Riethman, H. C., Moyzis, R. K., Meyne, J., Burke, D. T. and Olson, M. V., *Proc. Natl. Acad. Sci. USA*, **86**, 6240-6244 (1989).
213. Roberts, M. C., Pang, Y. J., Spencer, R. C., Winstanley, T. G., Brown, B. A. and Wallace, Jr., R. J. *Antimicrob. Agents Chemother.*, **35**, 2453-2455 (1991).

214. Rohozinski, J., Girton, L. E. and Van Etten, J. L., *Virology*, **168**, 363-369 (1989).
215. Romling, U., Grothues, D., Koopmann, U., Jahnke, B., Greipel, J. and Tummeler, B., *Abteilung Electrophoresis*, **13**, 646-648 (1992).
216. Roy, G., Wallenburg, J. C. and Chartrand, P., *Nucl. Acids Res.*, **16**, 768 (1988).
217. Rubin, C. M., Carrino, J. J., Dickler, M. N., Leibowitz, D., Smith, S. D. and Westbrook, C. A., *Proc. Natl. Acad. Sci. USA*, **85**, 2795-2799 (1988).
218. Russo, G., Isobe, M., Gatti, R., Finan, J., Batuman, O., Huebner, K., Nowell, P. C. and Croce, C. M., *Proc. Natl. Acad. Sci. USA*, **86**, 602-606 (1989).
219. Salama, S. M., Garcia, M. M. and Taylor, D. E., *Int. J. Syst. Bacteriol.*, **42**, 446-450 (1992).
220. Salama, S. M., Tabor, H., Richter, M. and Taylor, D. E., *J. Clin. Microbiol.*, **30**, 1982-1984 (1992).
221. Sanz, J. L., Marin, I., Ramirez, L., Abad, J. P., Smith, C. L. and Amils, R., *Nucl. Acids Res.*, **16**, 7827-7832 (1988).
222. Sarafis K. and Isaac-Renton J., *Am. J. Trop. Med. Hyg.*, **48**, 134-144 (1993).
223. Scherer, S. and Stevens, D. A., *Proc. Natl. Acad. Sci. USA*, **85**, 1452-1456 (1988).
224. Schlichting, C., Branger, C., Fournier, J. M., Witte W., Boutonnier, A., Wolz, C., Goullet, P. and Doring G., *J. Clin. Microbiol.*, **31**, 227-232 (1993).
225. Schmidt, M. C., Kao, C. C., Rui, P. and Berk, A. J., *Proc. Natl. Acad. Sci. USA*, **86**, 7785-7789 (1989).
226. Schoenline, P. V., Gallman, L. M. and Ely, B., *Gene*, **70**, 321-329 (1988).
227. Schoonmaker, D., Heimberger, T. and Birkhead, G., *J. Clin. Microbiol.*, **30**, 1491-1498 (1992).
228. Schwartz, D. C. and Cantor, C. R., *Cell*, **37**, 67-75 (1986).
229. Schwartz, D. C. and Koval, M., *Nature*, **338**, 520-522 (1989).
230. Schwartz, D. C., Saffran, W., Welsh, J., Haas, R., Goldenberg, M. and Cantor, C. R., *Cold Spring Harb. Symp. Quant. Biol.*, **47**, 189-195 (1983).
231. Searle, S., Campos, A. J., Coulson, M. R., Spithill, T. W. and Smith, D. F., *Nucl. Acids Res.*, **17**, 5081-5095 (1989).
232. Serwer, P., *Electrophoresis*, **8**, 301-304, (1987).
233. Serwer, P., *Applied and Theoret. Electrophor.*, **1**, 19-22 (1988).
234. Serwer, P. and Hayes, S. J., *Applied and Theoret. Electrophor.*, **1**, 95-98 (1989).
235. Serwer, P. and Hayes, S. J., *Biochem.*, **28**, 5827-5832 (1989).
236. Shayegani, M., Parsons, L. M., Waring, A. L., Donhowe, J., Goering, R., Archinal, W. A. and Linden, J., *J. Clin. Microbiol.*, **29**, 2768-2773 (1991).
237. Sheppard, M., Thompson, J. K., Anders, R. F., Kemp, D. J. and Lew, A. M., *Mol. Biochem. Parasitol.*, **34**, 45-52 (1989).
238. Shih, C.-K., Wagner, R., Feinstein, S., Kanik-Ennulat, C. and Neff, N., *Molec. and Cell. Biol.*, **8**, 3094-3103 (1988).
239. Simpson, A. M., Suyama, Y., Dewes, H., Campbell, D. A. and Simpson, L., *Nucl. Acids Res.*, **17**, 5427-5445 (1989).
240. Simske, J. S. and Scherer, S., *Nucl. Acids Res.*, **17**, 4359-4365 (1989).
241. Single L. A. and Martin D. R., *FEMS Microbiol. Let.*, **70**, 85-89 (1992).
242. Slater, G. W. and Noolandi, J., *Electrophoresis*, **10**, 413-428 (1989).
243. Smith, C. L., Econome, J. G., Schutt, A., Klco, S. and Cantor, C. R., *Science*, **236**, 1448-1453 (1987).
244. Smith, C. L., Klco, S. R. and Cantor, C. R., in *Genome Analysis: A Practical Approach*, Davies, K. E., ed., IRL Press, 41-72 (1988).
245. Smith, C. L., Matsumoto, T., Niwa, O., Klco, S., Fan, J., Yanagida, M. and Cantor, C., *Nucl. Acids Res.*, **15**, 4481-4489 (1987).
246. Smith, S. B., Aldridge, P. K. and Callis, J. B., *Science*, **243**, 203-243 (1989).
247. Snell, R. G. and Wilkins, R. J., *Nucl. Acids Res.*, **14**, 4401-4406 (1986).
248. Snewin, V. A., England, S. M., Sims, F. G. and Hyde, J. E., *Gene*, **76**, 41-52 (1989).
249. Sobral, B. W. and Atherly, A. G., *BioTechniques*, **7**, 938 (1989).
250. Sobral, B. W. and Atherly, A. G., *Nucl. Acids Res.*, **17**, 7359-7370 (1989).

251. Soldati, L. and Piffaretti, J. C., *Res. Microbiol.*, **142**, 489-498 (1991).
252. Sor, F., *Nucl. Acids Res.*, **16**, 4853-4863 (1988).
253. Southern, E. M., Anand, R., Brown, W. R. and Fletcher, D. S., *Nucl. Acids Res.*, **15**, 5925-5943 (1987).
254. Spitzer, E. D., Spitzer, S. G., Freundlich, L. F. and Casadevall, A., *Lancet*, **341**, 595-596 (1993).
255. Steel, P. E., Carle, G. F., Kobayashi, G. S. and Medoff, G., *Molec. and Cell. Biol.*, **9**, 983-987 (1989).
256. Steingrube, V. A., Wallace, Jr., R. J., Brown, B. A., Pang, Y., Zeluff, B., Steele, L. C. and Zhang, Y., *Antimicrob. Agents Chemother.*, **35**, 524-528 (1991).
257. Stellwagen, J. and Stellwagen, N. C., *Nucl. Acids Res.*, **17**, 1537-1548 (1989).
258. Stellwagen, N. C. and Stellwagen, J., *Electrophoresis*, **10**, 332-344 (1989).
259. Stewart, P. R., el-Adhami, W., Inglis, B. and Franklin, J. C., *J. Med. Microbiol.*, **38**, 270-277 (1993).
260. Storb, U., Haasch, D., Arp, B., Sanchez, P., Cazenave, P.-A. and Miller, J., *Molec. and Cell. Biol.*, **9**, 711-718 (1989).
261. Struelens, M. J., Deplano, A., Godard, C., Maes, N. and Serruys, E., *J. Clin. Microbiol.*, **30**, 2599-2605 (1992).
262. Sutherland, J. C., Emrick, A. B. and Trunk, J., *Electrophoresis*, **10**, 315-317 (1989).
263. Sutherland, J. C., Monteleone, D. C., Mugavero, J. H. and Trunk, J., *Anal. Biochem.*, **162**, 511-520 (1987).
264. Teow, W. L., Zaman, V., Ng, G. C., Chan, Y. C., Yap, E. H., Howe, J., Gopalakrishnakone, P. and Singh, M., *Int. J. Parasitol.*, **21**, 723-726 (1991).
265. Thal, L. A., Chow, J. W., Patterson, J. E., Perri, M. B., Donabedian, S., Clewell, D. B. and Zervos, M. J., *Antimicrob. Agents Chemother.*, **37**, 134-137 (1993).
266. Thiele, D. J., *Molec. and Cell. Biol.*, **8**, 2745-2752 (1988).
267. Traver, C. N., Klapholz, S., Hyman, R. W. and Davis, R. W., *Proc. Natl. Acad. Sci. USA*, **86**, 5898-5902 (1989).
268. Trent, R. J. and MacLeod, C., *Pathology*, **24**, 205-210 (1992).
269. Tulloch, D. L., Finch, L. R., Hillier, A. J., and Davidson, B. E., *J. Bacteriol.*, **173**, 2768-2775 (1991).
270. Turmel, C. and Lalande, M., *Nucl. Acids Res.*, **16**, 4727 (1988).
271. Upcraft, J. A., Boreham, P. F. and Upcraft, P., *Nucl. Acids Res.*, **17**, 3315 (1989).
272. U.S. Congress, Office of Technology Assessment, *Mapping our Genes- the genome projects, how big, how fast? OTA-BA-3*, Washington, DC. U.S. Government Printing Office (1988).
273. Van Daelen, R. A., Jonkers, J. J. and Zabel, P., *Plant Mol. Biol.*, **12**, 341-352 (1989).
274. Van der Blik, A. M., Baas, F., Ten Houte de Lange, T., Kooiman, P. M., Van der Velde-Koerts, T. and Borst, P., *EMBO J.*, **6**, 3325-3331 (1987).
275. Van der Blik, A. M., Lincke, C. R. and Borst, P., *Nucl. Acids Res.*, **16**, 4841-4851 (1988).
276. Van der Blik, A. M., Van der Velde-Koerts, T., Ling, V. and Borst, P., *Molec. and Cell. Biol.*, **6**, 1671-1678 (1986).
277. Van der Ploeg, L. H., Smith, C. L., Polvere, R. I. and Gottesdiener, K. M., *Nucl. Acids Res.*, **17**, 3217-3227 (1989).
278. Van Devanter, D. R., Trammell, H. M. and Von Hoff, D. D., *BioTechniques*, **7**, 143-144 (1989).
279. Van Dilla, M. A., Carrano, A. V., Christensen, M. L., de Jong, P. J., Grey, J., McNinch, J., Trask, B., van den Engh, G. and Yokobata, K., *Abstract #23: Human Genome I*, San Diego (1989).
280. Varma, A. and Kwon-Chung, K. J., *J. Clin. Microbiol.*, **30**, 2960-2967 (1992).
281. Vazquez, J. A., Beckley, A., Sobel, J. D. and Zervos, M. J., *J. Clin. Microbiol.*, **29**, 962-967 (1991).
282. Venter, U. and Horz, W., *Nucl. Acids Res.*, **17**, 1353-1368 (1989).
283. Ventra, L. and Weiss, A. S., *Gene*, **78**, 29-36 (1989).
284. Viovy, J. L., *Electrophoresis*, **10**, 429-441 (1989).
285. Vollrath, D. and Davis, R., *Nucl. Acids Res.*, **15**, 7865-7876 (1987).
286. Vollrath, D., Davis, R., Connelly, C. and Hieter, P., *Proc. Natl. Acad. Sci. USA*, **85**, 6027-6031 (1988).
287. Wallace, M. R., Fountain, J. W., Brereton, A. M. and Collins, F. S., *Nucl. Acids Res.*, **17**, 1665-1667 (1989).
288. Wanger, A. R., Morris, S. L., Ericsson, C., Singh, K. V., LaRocco, M. T., *J. Clin. Microbiol.*, **30**, 2583-2588 (1992).

289. Waterbury, P. F. and Lane, M. J., *Nucl. Acids Res.*, **15**, 3930 (1989).
290. Wei, M. Q. and Grubb, W. B., *J. Med. Microbiol.*, **37**, 187-191 (1992).
291. Wei, M. Q., Wang, F. U. and Grubb, W. B., *J. Med. Microbiol.*, **36**, 172-176 (1992).
292. Wellinger, R. J. and Zakian, V. A., *Proc. Natl. Acad. Sci. USA*, **86**, 973-977 (1989).
293. Wells, V. D., Wong, E. S., Murray, B. E., Coudron, P. E., Williams, D. S. and Markowitz, S. M., *Ann. Intern. Med.*, **116**, 285-292 (1992).
294. Wevrick, R. and Willard, H. F., *Proc. Natl. Acad. Sci. USA*, **86**, 9394-9398 (1989).
295. Widmer, A. F., Wenzel, R. P., Trilla, A., Bale, M. J., Jones, R. N. and Doebbeling, B. N., *Clin. Infect. Dis.*, **16**, 372-376 (1993).
296. Wilson, M. R. and Coussens, P. M., *Virology*, **185**, 673-680 (1991).
297. Woolf, T., Lai, E., Kroneberg, M. and Hood, L., *Nucl. Acids Res.*, **16**, 3863-3874 (1988).
298. Wu, C.-I., Lyttle, T. W., Wu, M.-L. and Lin, G.-F., *Cell*, **54**, 179-189 (1988).
299. Yan, W., Chang, N. and Taylor, D. E., *J. Infect. Dis.*, **163**, 1068-1072 (1991).
300. Zakian, V. A. and Blanton, H. M., *Molec. and Cell. Biol.*, **8**, 2257-2260 (1988).
301. Zakian, V., Blanton, H., Liebchen, W. and Dani, G., *Molec. and Cell. Biol.*, **6**, 925-932 (1986).
302. Zhang, Y., Mazurek, G. H., Cave, M. D., Eisenach, K. D., Pang, Y., Murphy, D. T. and Wallace, Jr., R. J., *J. Clin. Microbiol.*, **30**, 1551-1556 (1992).
303. Zingler, G., Ott, M., Blum, G., Falkenhagen, U., Naumann, G., Sokolowska-Kohler, W. and Hacker, J., *Microb. Pathog.*, **12**, 299-310, (1992).
304. CHEF (US Patent 5,165,898 issued to Stanford University) is licensed to Bio-Rad Laboratories.
305. PACE (US Patent 5,084,157 issued to California Institute of Technology) is exclusively licensed to Bio-Rad Laboratories.

Section 7

Appendix

7.1 Screen Mode Program

The screen mode program in the GenePath system is used to quickly check your sample plugs for DNA concentration and for preliminary analysis of isolates. The screen mode **can not** be used with the Group 4 Reagent Kit (*Candida*). The screen mode takes 4 hours to complete, so it can be run during the day. To increase the migration rate of the DNA fragments, the screen mode uses a higher voltage gradient, and a more dilute electrophoresis buffer and agarose concentration. It is **highly recommended** that the appropriate organism program be run after the screen mode program. Slight differences in banding pattern between isolates may be hard to visualize with the screen mode program.

The screen mode requires a different buffer and agarose concentration. To prepare the buffer and agarose for the screen mode program, follow the protocol below. For a more detailed procedure, refer to the Group Reagent Kit manuals.

Agarose Preparation

1. Melt a 100 ml bottle of agarose in a microwave oven. This will take 1 to 2 minutes. Check the agarose every 30 seconds to make sure the agarose does not boil over.
2. Carefully remove 30 ml of agarose from the bottle. Add 30 ml of water and mix by swirling or with a stir bar.
3. Pour the agarose into the casting stand. Let the agarose solidify for 30 to 60 minutes at room temperature.

Note: This gel is fragile, so care should be taken whenever handling the gel.

Buffer Preparation

1. Add 50 ml of the Electrophoresis Running Buffer (1/2 bottle) to 1950 ml deionized water. Mix the buffer and add to the gel chamber.

7.2 Download Program

The download program is used to download new programs, as they become available from Bio-Rad, from a computer through a RS-232 cable. Once the new programs are downloaded to the GenePath system, they will be displayed in the menu along with the other programs.

7.3 Systems, Accessories, and Reagents for Pulsed Field Electrophoresis

Catalog

Number	Product Description
3 10-0000	GenePath Strain Typing System , 120 VAC, includes GenePath Electrophoresis System , Electrophoresis Cell; Cooling Module; variable speed pump; Tygon tubing, 12 feet; 14 cm wide x 13 cm long casting stand and frame; 10 well comb and comb holder; 50 well disposable sample plug mold; leveling bubble; cables; 3/8 inch straight tubing connectors, 2; 0.5 A FB fuses, 2; <i>S. cerevisiae</i> DNA size standards; Pulsed Field Certified Agarose sample, 5 g; Chromosomal Grade Agarose sample, 5 g; manual. Also includes Standard Documentation System , including Mini-Transilluminator, DS-34 Camera, Standard Electrophoresis Hood, Deep Yellow Camera Filter.
310-0001	GenePath Strain Typing System , 220/240 VAC
3 1 0-0008	GenePath Strain Typing System , 100 VAC
310-0015	GenePath Electrophoresis System , 100 VAC, includes Electrophoresis Cell; Cooling Module; variable speed pump; Tygon tubing, 12 feet; 14 cm wide x 13 cm long casting stand and frame; 10 well comb and comb holder; 50 well disposable sample plug mold; leveling bubble; cables; 3/8 inch straight tubing connectors, 2; 0.5 A FB fuses, 2; <i>S. cerevisiae</i> DNA size standards; Pulsed Field Certified Agarose sample, 5 g; Chromosomal Grade Agarose sample, 5 g; manual.
310-0016	GenePath Electrophoresis System , 120 VAC
310-0017	GenePath Electrophoresis System , 220/240 VAC
3 1 0-0002	GenePath Group 1 Reagent Kit
31 0-0003	GenePath Group 2 Reagent Kit
3 1 0-0004	GenePath Group 3 Reagent Kit
3 1 0-0005	GenePath Group 4 Reagent Kit
3 1 0-0007	GenePath Gel Kit
170-3742	Standard Documentation System , 120 VAC, includes Mini-Transilluminator, 100 V, DS-34 camera, standard electrophoresis hood, Deep Yellow DS-34 camera filter
170-3746	Standard Documentation System , 100 VAC
170-3747	Standard Documentation System , 220/240 VAC
170-3743	Wide/Long Documentation System , 120 VAC, includes Mini-Transilluminator, 100 V, DS-34 camera, wide/long electrophoresis hood, Deep Yellow DS-34 camera filter
170-3748	Wide/Long Documentation System , 100 VAC
170-3749	Wide/Long Documentation System , 220/240 VAC

Catalog Number	Product Description
170-3745	Mini-Transilluminator, 100 VAC
170-3737	Mini-Transilluminator, 120 VAC
170-3738	Mini-Transilluminator, 220/240 VAC
170-3739	Standard Electrophoresis Hood, 5" x 7" (12.7 x 17.8 cm)
170-3740	Wide/Long Electrophoresis Hood, 6" x 9" (15.2 x 22.9 cm)
170-3741	DS-34 Camera
170-3744	Deep Yellow DS-34 Camera Filter
170-3643	Gel-Scoop, for transporting gels
170-3751	Mini-Transilluminator Fluorescent Lamp, 302 nm
223-6280	Microfuge
196-1016	Water Bath
223-9480	1.5 ml Microfuge tube (not pre-sterilized)
223-9430	2.0 ml Microfuge tube (not pre-sterilized)
223-9550	Disposable Transfer Pipet
223-9309	Pipet tip: I-I 00 µl
223-9372	Pipettip: 100-1000 µl
224-4005	Pipet: 2-20 µl
224-4010	Pipet: IO-100 µl
224-4020	Pipet: 100-1000 µl
170-4046	Leveling Table
224-0110	1.5 ml Screwcap tube, sterile
224-0140	2.0 ml Screwcap tube, sterile
223-9643	5 ml sterile culture tubes
223-9950	1.5 ml Spectrophotometer cuvettes
223-9955	3.5 ml Spectrophotometer cuvettes



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